Beyond Dodge Rate: Testing Humanoid Dodgeball Under Camera Failures Supplementary Material

Kingston Kuan[1] Samuel J Chew[2] Khai Loong Aw[3] Brandon Ong[1]
[1]Hebbian Robotics [2]Carnegie Mellon University [3]Stanford University

https://hebbian-robotics.github.io/beyond-dodge-rate/

APPENDIX A
EVALUATION PROTOCOL

This supplement accompanies *Beyond Dodge Rate: Testing Humanoid Dodgeball Under Camera Failures*. Section A gives reproducibility details, Section B reports supporting results, and Section C describes the demo and releases.

A. Policies, simulator, and task

The controller combines frozen SONIC and Theia bases with depth adaptation trained by PPO through RSL-RL in mjlab/MuJoCo Warp, using the sources cited in the main paper. π_1 is the released iteration-4000 model. The saved configuration for π_2 matches the released reference: training seed 42, 4,000 updates, 512 environments, and about 49.15 million transitions, distributed across eight GPUs in our run. The weights differ. Initialization, random-number sequences, and resume histories are incomplete, so these are not a controlled comparison of training methods or independent training replications with fully controlled histories.

Evaluation fixes weights, native gains, torque limits, depth noise, active-ball collisions, and fall checks. MuJoCo 3.3.7 and ONNX Runtime 1.23.2 run on CPU, with 50 Hz control and 25 Hz depth frames. All trials correct the inactive-ball bug by disabling collisions while the ball is parked below the floor and restoring them at launch.

The ball launches at 1 s. The native generator aims at one of six body locations, adds random position offsets and $\pm 30^\circ$ direction variation, and uses a nominal flight time of 0.55–0.62 s. Trials end at 4 s or on a fall. The allowed area is a circle of radius 1 m centred on the pelvis starting position. A fall occurs when pelvis height is below 0.45 m or upright-axis cosine is below 0.4 (about 66.4° tilt).

B. Cohorts and fixed comparisons

The input-handling experiment fixes 192 seeds (7300–7491) per policy before observing outcomes. Each seed runs healthy input, continuous stale input, and continuous blank input: 1,152 trials across both policies. Fault onset is 160 ms after launch (1.16 s), and faults last until trial end.

The original fresh-image restoration experiment uses a separate exploratory 16-seed pilot per policy (8000–8015), then 64 new seeds (9000–9063). Each seed runs five conditions: healthy input, continuous stale input, continuous blank input, stale input followed by fresh-image restoration at 2 s, and blank input followed by restoration at 2 s. Faults begin at 1.16 s (160 ms after launch). This gives 160 pilot and 640 original restoration trials. After the pilot, we fixed the new seeds and the primary contrast: restoration versus continuous stale input for π_1. The other policy and blank-input contrasts are secondary. Pilot outcomes are excluded.

Restoration follows a fixed schedule rather than being triggered by throw resolution. All conditions retain the four-second horizon, subject to the native fall termination rule, without further outcome-dependent stopping. Histories are not reset. All stale-input throws resolve by restoration within numerical tolerance. The original 2,496-trial package also includes exploratory cohorts beyond the 192-seed input-handling experiment. Exploratory controls and post-hoc scoring analyses are not pooled into the original restoration estimate.

C. Matching and shared ball launches

Within each policy and seed, prelaunch robot positions and velocities, launch time, target, ball position and velocity, and flight time must match across camera conditions. Camera schedules are checked. Missing or mismatched trials halt analysis. Restoration pairs also have identical recorded physical samples before 2 s. Different robot motion before launch can change the simulator's aim, so equal seeds need not yield equal throws across policies. Comparisons concern camera inputs within each policy rather than a policy ranking.

The shared-throw control tests whether responses persist under matched ball launches. For index $i = 0, \ldots, 63$, donor seed $9000+i$ supplies a healthy-input throw from π_1 when i is even and π_2 when i is odd. This selects 32 launches per policy without filtering outcomes. Simulation seeds 10000–10063 apply each launch to both policies under all five restoration conditions, with the same 1.16 s fault onset and 2 s restoration schedule, giving 640 trials. Twenty separate preflight trials use two other simulation seeds and are retained in the shared-throw archive.

At launch, the evaluator retains native collision settings, outcome accounting, and random draws, then replaces only the ball's initial position and velocity before integration. Exact throw parameters match across all ten policy-condition

combinations per seed. Robot states can still differ. The target label describes the donor's original aim, not the receiving robot's body part. Subsequent ball trajectories are simulated, not imposed. This control reuses recorded launch conditions and adds no independent training runs.

D. Scoring, timing, and uncertainty

Task success requires a dodge, no fall, and area compliance throughout the trial. A dodge is a simulator-recorded miss. Trials ending before throw resolution (aborted throws) count as failures and remain in the denominator. Area compliance fails at the first sampled exit, even if the robot returns. The area criterion measures pelvis containment, not whole-body containment or damage, and is neither a court rule nor an injury threshold.

The post-hoc timing analysis reuses all 1,152 input-handling trials without changing seeds or outcome labels. A hit resolves at native contact. A miss resolves at the first 20 ms control step at least one second after launch if the ball remains active after contact processing and the fall check. Resolution is a scoring event, not a measurement of closest approach. An area exit is late only if it occurs strictly after resolution; ties count as at-or-before resolution. Area exits use 50 Hz samples without interpolation.

Paired 95% confidence intervals use 10,000 bootstrap resamples of whole evaluation seeds, retaining their conditions together. Shared-throw comparisons retain both policies in each resampled seed. The input-handling, original restoration, shared-throw input-effect, and stale-to-blank analyses use random seed 20261014. The area-radius/duration analysis uses 20261015, and its continuous-distance analysis uses 20261016. Intervals are nominal per comparison, without multiplicity adjustment. Individual-condition intervals are Wilson intervals. Evaluation seeds are not independent training replications.

The trial archives retain configurations, camera schedules, physical traces, and outcome flags. Analysis code computes intervals and validates matching from these records. These records do not by themselves imply identical replay across different execution environments.

E. Controller and station reward

Each 25 Hz depth frame is encoded by Theia. A perception module aggregates 16 recent visual tokens into a feature. At 50 Hz, the actor combines this feature with a standing reference and ten samples each of gyroscope readings, joint-position offsets from the default pose, joint velocities, previous actions, and projected gravity. The standing reference includes default joint positions, zero joint velocities, and orientation relative to the reference. The 29 normalized outputs specify joint targets through native scaling, PD gains, and torque limits.

Fault injection replaces depth values before preprocessing and Theia encoding. Visual and proprioceptive histories keep updating through switches. Old visual tokens remain until newer tokens replace them. Blank input supplies zero-valued depth frames at the native camera rate. Fresh-image restoration resumes the native noisy camera stream.

The pinned training environment inherits the station term

$$r_{\text{station}} = \exp\left(-\frac{\|\mathbf{p}_{\text{ref}} - \mathbf{p}_{\text{robot}}\|_2^2}{\sigma^2}\right). \tag{1}$$

The weight is $w = 0.3$ and the scale is $\sigma = 1$ m. The positions are the reference and robot anchor positions in world coordinates, including height. The reward manager applies the weight and its time-step scaling. The term is `motion_global_anchor_position_error_exp` in mjlab 1.4.0. The environment also rewards ball clearance and penalizes action changes and joint-limit violations. The depth-adaptation wrapper adds hit and fall penalties. The horizontal, trial-wide area criterion is additional to this smooth reward. Its 1 m scale supplies no task-level justification for a hard area radius. Training hyperparameters were fixed, so their sensitivity is outside these tests.

Appendix B
Detailed Results

A. Input handling and late area exits

Blank versus stale input raises π_1 success from 70/192 to 171/192 (+52.6 percentage points; paired 95% confidence interval +43.8 to +60.9). For π_2, success falls from 158/192 to 100/192 (-30.2 percentage points; confidence interval -37.0 to -23.4). For π_1, area compliance rises from 70/192 to 181/192 while dodges fall from 191/192 to 182/192. For π_2, area compliance rises from 182/192 to 190/192 while dodges fall from 165/192 to 101/192. There are no falls. All conditions, including healthy input, are reported in main-paper Table III.

Under stale input, all 122 π_1 area exits occur after throw resolution; 121 follow a dodge. Its 11 exits under blank input are also late. For π_2, stale input produces seven exits at or before resolution and three afterward. Its two exits under blank input are late. These timings describe when area-compliance failures occur; they do not explain the learned policies.

B. Original fresh-image restoration experiment

For π_1, restoration versus continuous stale input changes success from 32/64 to 61/64 (+45.3 percentage points; paired 95% confidence interval +32.8 to +57.8). There are 29 trials changing from failure to success and none changing from success to failure. For π_2, success changes from 53/64 to 51/64 (-3.1 percentage points; confidence interval -7.8 to 0), with no trials changing from failure to success and two changing from success to failure. Both π_1 conditions dodge 64/64 throws, and both π_2 conditions dodge 54/64. All five conditions have zero falls and aborted throws. Main-paper Table II gives all counts, including the smaller or absent effects of restoration after blank input. These results concern area compliance after the encounter and establish neither a neural mechanism nor a universal fallback.

TABLE I
SHARED-THROW CONTROL. COUNTS OUT OF 64 PER CONDITION; FALLS AND ABORTED THROWS ARE ZERO. FRESH-IMAGE RESTORATION OCCURS AT 2 S. INITIAL BALL POSITIONS AND VELOCITIES MATCH ACROSS POLICIES; ROBOT STATES CAN DIFFER.

Policy	Camera input	Dodge	In area	Success
π_1	Healthy	61/64	64/64	61/64
	Stale	60/64	27/64	23/64
	Stale + recovery	60/64	64/64	60/64
	Blank	57/64	60/64	53/64
	Blank + recovery	57/64	62/64	55/64
π_2	Healthy	51/64	60/64	47/64
	Stale	52/64	58/64	46/64
	Stale + recovery	52/64	57/64	45/64
	Blank	41/64	62/64	39/64
	Blank + recovery	41/64	61/64	38/64

C. Area radius, duration, and continuous distance

Post-hoc analyses use area radii 0.5, 0.75, 1.0, 1.25, 1.5, and 2.0 m and observation durations 2.0, 2.5, 3.0, and 4.0 s in the original restoration and shared-throw cohorts. Increases in success rate emerge mainly late and shrink with larger area radii. At four seconds, increasing the radius from 1 to 1.5 m reduces the additional successful π_1 trials from 29/64 to 2/64 in the original cohort and from 37/64 to 1/64 in the shared-throw control.

Pelvis distance is horizontal distance from its starting position. Maximum pelvis distance is its largest value during the trial; final pelvis distance is its value at trial end. These summaries include all 64 seeds per policy in each cohort, including hits, and reuse recorded trials. They do not apply an area criterion.

For π_1, restoration reduces mean maximum pelvis distance from 1.043 to 0.860 m (paired difference -0.184 m; 95% confidence interval -0.226 to -0.146 m). The corresponding shared-throw difference is -0.179 m. Mean final pelvis distance changes by -0.608 m in the original cohort and -0.581 m in the shared-throw cohort.

For π_2, the mean maximum pelvis distance changes by $+0.010$ m in the original cohort (paired 95% confidence interval $+0.005$ to $+0.018$ m) and -0.002 m under shared throws (interval -0.019 to $+0.012$ m). The latter interval spans zero. Mean final pelvis distance changes by -0.070 and -0.060 m, respectively. Thus, final distance can decrease without a corresponding decrease in maximum distance.

D. Shared-throw control

Blank versus stale input changes success by +46.9 percentage points for π_1 (paired 95% confidence interval +31.2 to +62.5) and -10.9 for π_2 (interval -21.9 to 0). The between-policy difference is +57.8 percentage points (interval +40.6 to +75.0). The π_2 interval reaches zero, so this control alone does not establish a harmful blank-input effect for that policy.

Fresh-image restoration raises π_1 success from 23/64 to 60/64, with 60/64 dodges in both conditions. For π_2, success changes from 46/64 to 45/64, with 52/64 dodges in both. No falls or aborted throws occur. These response differences persist under matched ball launches, although robot states, weights, and incompletely documented training histories can differ.

E. Stale-to-blank control

The control reruns seeds 9000–9063 for both policies in three conditions, giving 384 trials. All conditions receive fresh frames before 1.16 s, then repeat the last pre-fault frame. At the fixed 2 s switch, they continue stale input, supply blank input, or resume fresh frames. Histories keep updating. All 128 triplets pass launch-state, throw-record, supplied-frame hash checks before 2 s, and recorded pelvis-trace checks through 2 s. Native camera frames at the switch also match before substitution. All throws resolve by the switch within numerical tolerance. Every condition retains every seed through the four-second horizon.

For π_1, success is 31/64 with continued stale input, 62/64 with blank input, and 62/64 with fresh frames. The two switches change the same 31 trials from failure to success, with none worsening. Each success-rate increase is $31/64 = 48.4$ percentage points (paired 95% confidence interval 35.9–60.9). For π_2, all three conditions succeed on the same 54/64 trials. All π_1 trials dodge; each π_2 condition dodges 55/64. No falls or aborted throws occur.

Fresh-image restoration versus blank input yields zero discordant success pairs in 64 trials per policy. Under independent evaluation-seed sampling, the one-sided exact 95% upper bound on discordance probability p follows from $(1-p)^{64} = 0.05$, giving $p \leq 1 - 0.05^{1/64} = 0.0457$, or 4.6% rounded. These are separate per-policy bounds. The empirical paired bootstrap has zero width for these identical binary outcomes, so it cannot establish a population equivalence margin.

Fresh frames reduce mean final pelvis distance relative to blank input by 0.295 m for π_1 (paired 95% confidence interval 0.253–0.341 m) and 0.117 m for π_2 (0.091–0.144 m). The fresh-minus-blank difference in mean maximum pelvis distance is $+0.0015$ m for π_1 (interval -0.0008 to $+0.0039$ m) and -0.0046 m for π_2 (-0.0124 to $+0.0040$ m). Blank input with retained history reproduces the observed success outcomes, while fresh frames produce smaller mean final distances.

The simulator, robot, and weight files match the frozen runtime manifests. The local rerun uses Python 3.12.3, NumPy 2.2.6, MuJoCo 3.3.7, ONNX Runtime 1.23.2 CPU inference with one thread per session, and an Intel Core Ultra 9 275HX. Trajectories differ from the published archives: continued-stale/fresh success changes from archived 32/61 to local 31/62 for π_1, and from 53/51 to 54/54 for π_2, all out of 64. Every reported switch comparison uses the local triplets. The cause of the cross-runtime differences remains unresolved. This exploratory control reuses the original restoration seed IDs and is analysed separately.

TABLE II
STALE-TO-BLANK CONTROL. COUNTS OUT OF 64; FALLS AND ABORTED THROWS ARE ZERO. MEAN MAXIMUM AND FINAL PELVIS DISTANCES ARE IN METRES AND INCLUDE EVERY TRIAL, INCLUDING HITS. ALL TRIALS REACH FOUR SECONDS.

Policy	Input after 2 s	Dodged	Area-compliant	Successful	Mean maximum	Mean final
π_1	Stale	64	31	31	1.040	1.020
	Blank	64	62	62	0.855	0.724
	Fresh	64	62	62	0.856	0.429
π_2	Stale	55	59	54	0.713	0.585
	Blank	55	59	54	0.726	0.652
	Fresh	55	59	54	0.721	0.534

APPENDIX C
DEMO: CAMERA FAILURES IN SIMULATED HUMANOID DODGEBALL

A. What viewers see

The demo presents synchronized replays of the experiments in this paper. Viewers compare camera conditions, pause at measured events, and follow the robot's pelvis relative to the allowed area of radius 1 m. The examples use the first declared seeds; aggregate results are reported separately so that an illustrative trial is not mistaken for average performance.

The overview video first compares healthy input with stale-input faults at ball launch and 320 ms later. In the illustrated trial, the later stale-input fault allows a dodge but the robot subsequently leaves the area. The video then compares fresh-image restoration, followed by healthy, stale, and blank input for π_2. The opening slide defines π_1 and π_2 and identifies their sources; separate tables report the 64-seed original restoration experiment and the 192-seed input comparison. These are two sets of learned weights trained with the same method and recorded settings, not two proposed training methods.

The recovery replay compares continuous stale input with fresh frames restored at two or three seconds. All three robots dodge the same throw; only the two-second restoration stays within the area in this example. The three-second example is labeled exploratory. The paper's aggregate restoration results, rather than this single replay, support the reported effect.

B. Playback and setup

The videos use aligned views and synchronized holds so that the encounter and later movement can be compared. An orange marker remains briefly after the ball's native removal to make its last recorded position visible. The marker is a display aid, not a simulated rebound or continued ball physics.

The demo consists of recorded simulation, rather than live policy execution. A laptop with a browser is sufficient for presentation. The presenters bring the laptop; a power outlet and an optional external display are useful. Downloaded videos play offline. No physical robot, GPU, or special floor area is needed. Online playback of the YouTube version requires internet access.

C. Video

The overview video is available at https://youtu.be/iXgu4AaxdYE. Synchronized replays are available on the project website linked below the paper title.

The results concern two fixed policies in one simulated G1 task. They do not establish hardware safety or variation across independent training runs.

D. Code, data and weight availability

Evaluation code, original configurations, dependency locks, and provenance are available through the project website. Four original recorded-trial archives are publicly released at https://huggingface.co/datasets/hebbianrobotics/beyond-dodge-rate. They cover the original 2,496-trial study, the 800 pilot and original restoration trials, the 660 shared-throw and preflight trials, and the 128 later-restoration trials. Repository manifests specify archive and file hashes.

The released reference weights are available at https://huggingface.co/nepyope/g1_depth_dodge. The evaluated π_2 actor and perception ONNX files are public at https://huggingface.co/hebbianrobotics/g1-depth-dodge. The shared encoders come from the reference release. The retained π_2 training checkpoint is not distributed. Fresh simulation requires the pinned simulator, robot assets, and applicable model files under their original terms. Recomputing the original statistics requires the trial archives. Cross-runtime exact replay remains a separate qualification step.

The 384 stale-to-blank control records await public release and are separate from the four public archives. Their archive and record hashes are in `data/stale-to-blank-manifest.json`; the retained archive is named `stale-to-blank-data-v1.tar.gz`. It contains each condition's configuration, frame-hash ledger, physical traces, native outcomes, and execution metadata. Reference summaries and per-seed outcomes are in `results/stale-to-blank-v1/`.

E. Reproducing the stale-to-blank analysis

The repository includes `scripts/analyze_stale_to_blank.py`, its helper modules, both frozen configurations under `configs/`, the data manifests, and locked dependencies. Use Python 3.12 and `uv`. Download `camera-recovery-data-v1.tar.gz` from the dataset above and, once obtained, place the separate `stale-to-blank-data-v1.tar.gz` alongside it in `data/`. The latter archive is required for reproducing the retained control results and awaits public release. Run the following from the repository root; the trial and output directories must not already exist.

```
uv sync --locked
mkdir -p outputs/stale-to-blank-records
tar -xzf data/stale-to-blank-data-v1.tar.gz \
  -C outputs/stale-to-blank-records
uv run python scripts/analyze_stale_to_blank.py \
  --trials outputs/stale-to-blank-records \
  --recovery-data \
    data/camera-recovery-data-v1.tar.gz \
  --output outputs/stale-to-blank-analysis
```

The command verifies the recovery archive and its file hashes, recomputes all 384 control outcomes, checks all 128 matched triplets and their pre-switch camera and physical histories, and repeats the 10,000-resample paired bootstrap. It writes `summary.json` and `trials.csv` under `outputs/stale-to-blank-analysis/`; these should match the supplied files in `results/stale-to-blank-v1/`. It also generates the supplementary `switch-table.tex`. The original archive supplies the separate historical comparison, not pooled observations. This recorded-data analysis needs no weights, simulator execution, or GPU and does not assert exact cross-runtime replay. To run fresh simulations, use `scripts/evaluate.py` with `configs/stale-to-blank-upstream-v1.json` and `configs/stale-to-blank-team-v1.json` and the corresponding corrected simulator inputs and weights. The repository's reproduction guide gives the commands; fresh runs remain separate from the retained results.